Daniel Fried

Contact Information

dfried@cs.cmu.edu
dpfried.github.io

Positions

Assistant Professor, Carnegie Mellon University — 2022 – present
Language Technologies Institute, School of Computer Science

Education

UC Berkeley — 2015 – 2021
Ph.D. in Computer Science
Adviser: Dan Klein
Thesis: Learning Grounded Pragmatic Communication

University of Cambridge, Churchill College — 2014 – 2015
M.Phil. in Computer Science, *with distinction*
Adviser: Stephen Clark
Thesis: Low Rank Tensor Approximations for Compositional Distributional Semantics

University of Arizona — 2010 – 2014
B.S. in Computer Science, Mathematics, and Information Science, *summa cum laude*
Thesis Adviser: Mihai Surdeanu
Thesis: Predicting Community Traits Using the Language of Food on Social Media

Former Positions

Research Scientist, Meta — 2024 – 2026
Areas: LLM-based coding agents

Research Consultant, GitHub Next — 2023 – 2024
Areas: LLM-based models for code editing

Postdoc, University of Washington — 2021 – 2022
Host: Luke Zettlemoyer

Visiting Researcher, Facebook AI Research — 2021 – 2022
Host: Mike Lewis

Research Intern, Google DeepMind — Summer 2019
Hosts: Aida Nematzadeh, Stephen Clark, Chris Dyer
Project: Structured models for language-conditioned video segmentation

Research Intern, Microsoft Research — Summer 2016
Hosts: Hoifung Poon, Chris Quirk, Kristina Toutanova, Scott Wen-Tau Yih
Project: Learning to rank for personalized medicine

Undergraduate Research Assistant, University of Arizona — 2012 – 2014
Advisers: Mihai Surdeanu, Stephen Kobourov, Paul R. Cohen
Areas: Question answering, language grounding, data visualization

Research Intern, Nara Institute of Science and Technology — Fall 2013
Host: Kevin Duh
Project: Incorporating relational semantics in word embeddings

Research Intern, RWTH Aachen University — Summer 2013
Hosts: Ali Jannesari, Zhen Li
Project: Supervised learning for automatic code parallelization

Engineering Intern, Microsoft — Summer 2012
Project: Data warehousing, analytics, and visualization for Xbox Live

Honors & Awards

Schmidt Sciences AI 2050 Early Career Fellow	2026
NSF CAREER Award	2026
Microsoft Research Faculty Fellowship	2026
Okawa Research Award	2023
Google Ph.D. Fellowship in Natural Language Processing	2019 – 2021
Outstanding Graduate Student Instructor, UC Berkeley	2018
Outstanding Reviewer, ACL	2018, 2020, 2021, 2022
Outstanding Reviewer, NeurIPS	2019
Best M.Phil. Student Award, Cambridge Computer Laboratory	2015
Churchill Scholarship	2014 – 2015
NDSEG Fellowship	2014
Finalist, Hertz Graduate Fellowship	2014
Outstanding Senior Award in Research, U. Arizona College of Science	2014
Outstanding Senior Award in Academics, U. Arizona Computer Science	2014
Outstanding Senior Award in Academics, U. Arizona Information Science	2014
Barry M. Goldwater Scholarship	2013
National Merit Scholar	2010 – 2014
Flinn Scholarship, Flinn Foundation of Arizona	2010 – 2014
Presidential Scholar, U.S. Department of Education	2010

Journal Publications

1. **Tree Search for Language Model Agents**
   Jing Yu Koh, Stephen McAleer, Daniel Fried, and Ruslan Salakhutdinov
   Transactions on Machine Learning Research (TMLR), 2025
2. **StarCoder: May the Source Be With You!**
   Raymond Li et al. (68 authors from the BigCode Project)
   Transactions on Machine Learning Research (TMLR), 2023
3. **Human-Level Play in the Game of Diplomacy by Combining Language Models with Strategic Reasoning**
   FAIR Diplomacy Team
   Science, 2022
4. **Syntactic Structure Distillation Pretraining for Bidirectional Encoders**
   Adhiguna Kuncoro*, Lingpeng Kong*, Daniel Fried*, Dani Yogatama, Laura Rimell, Chris Dyer, and Phil Blunsom
   Transactions of the Association for Computational Linguistics (TACL), 2020
5. **Higher-Order Lexical Semantic Models for Non-Factoid Answer Reranking**
   Daniel Fried, Peter Jansen, Gustave Hahn-Powell, Mihai Surdeanu, and Peter Clark
   Transactions of the Association for Computational Linguistics (TACL), 2015

Conference Publications

6. **Reasoning with Latent Tokens in Diffusion Language Models**
   Andre He, Sean Welleck*, Daniel Fried*
   NeurIPS, 2026. **Spotlight (top 1% of submissions)**
7. **Odysseys: Benchmarking Web Agents on Realistic Long Horizon Tasks**
   Lawrence Jang*, Jing Yu Koh*, Daniel Fried, Ruslan Salakhutdinov
   Conference on Language Modeling (COLM), 2026
8. **MetaLint: Generalizable Idiomatic Code Quality Analysis through Instruction-Following and Easy-to-Hard Generalization**
   Atharva Naik, Lawanya Baghel, Dhakshin Govindarajan, Darsh Agrawal, Yiqing Xie, Daniel Fried, Carolyn Rose
   Conference on Language Modeling (COLM), 2026
9. **Agent Psychometrics: Task-Level Performance Prediction in Agentic Coding Benchmarks**
   Chris Ge, Daria Kryvosheieva, Daniel Fried, Uzay Girit, Kaivalya Hariharan
   Conference on Language Modeling (COLM), 2026

10. **Scaling Test-Time Compute for Agentic Coding**
Joongwon Kim, Wannan Yang, Kelvin Niu, Hongming Zhang, Yun Zhu, Eryk Helenowski, Ruan Silva, Zhengxing Chen, Srinivasan Iyer, Manzil Zaheer, Daniel Fried, Hannaneh Hajishirzi, Sanjeev Arora, Gabriel Synnaeve, Ruslan Salakhutdinov, Anirudh Goyal
Conference on Language Modeling (COLM), 2026

11. **Hybrid-Gym: Training Coding Agents to Generalize Across Tasks**
Yiqing Xie, Emmy Liu, Gaokai Zhang, Nachiket Kotalwar, Shubham Gandhi, Sathwik Acharya, Xingyao Wang, Carolyn Rose, Graham Neubig, Daniel Fried
International Conference on Machine Learning (ICML), 2026

12. **Propose, Solve, Verify: Self-Play Through Formal Verification**
Alex Wilf, Pranjal Aggarwal, Bryan Parno, Daniel Fried, Louis-Philippe Morency, Paul Pu Liang, Sean Welleck
International Conference on Machine Learning (ICML), 2026

13. **Toward Training Superintelligent Software Agents through Self-Play SWE-RL**
Yuxiang Wei, Zhiqing Sun, Emily McMilin, Jonas Gehring, David Zhang, Gabriel Synnaeve, Daniel Fried, Lingming Zhang, Sida Wang
International Conference on Machine Learning (ICML), 2026

14. **Success and Cost Elicit Convention Formation for Efficient Communication**
Saujas Vaduguru, Yilun Hua, Yoav Artzi, Daniel Fried
Annual Meeting of the Association for Computational Linguistics (ACL), 2026

15. **Generative Value Conflicts Reveal LLM Priorities**
Andy Liu, Kshitish Ghate, Mona Diab*, Daniel Fried*, Atoosa Kasirzadeh*, Max Kleiman-Weiner*
International Conference on Learning Representations (ICLR), 2026

16. **From Reproduction to Replication: Evaluating Research Agents with Progressive Code Masking**
Gyeongwon James Kim, Alex Wilf, Louis-Philippe Morency, Daniel Fried
International Conference on Learning Representations (ICLR), 2026

17. **SWE-RL: Advancing LLM Reasoning via Reinforcement Learning on Open Software Evolution**
Yuxiang Wei, Olivier Duchenne, Jade Copet, Quentin Carbonneaux, Lingming Zhang, Daniel Fried, Gabriel Synnaeve, Rishabh Singh, and Sida I. Wang
Neural Information Processing Systems (NeurIPS), 2025

18. **Identifying and Interactively Refining Ambiguous User Goals for Data Visualization Code Generation**
Mert Inan, Anthony Sicilia, Alex Xie, Saujas Vaduguru, Daniel Fried, and Malihe Alikhani
Empirical Methods in Natural Language Processing (EMNLP), 2025

19. **Rewarding the Unlikely: Lifting GRPO Beyond Distribution Sharpening**
Andre He, Daniel Fried, and Sean Welleck
Empirical Methods in Natural Language Processing (EMNLP), 2025

20. **mrCAD: Multimodal Refinement of Computer-aided Designs**
William P. McCarthy, Saujas Vaduguru, Karl D. D. Willis, Justin Matejka, Judith E. Fan, Daniel Fried, and Yewen Pu
Findings of EMNLP, 2025

21. **Inducing Programmatic Skills for Agentic Tasks**
Zora Zhiruo Wang, Apurva Gandhi, Graham Neubig, Daniel Fried
Conference on Language Modeling (COLM), 2025

22. **RepoST: Scalable Repository-Level Coding Environment Construction with Sandbox Testing**
Yiqing Xie, Alex Xie, Divyanshu Sheth, Pengfei Liu, Daniel Fried, Carolyn Rose
Conference on Language Modeling (COLM), 2025

23. **Improving Model Factuality with Fine-grained Critique-based Evaluator**
Yiqing Xie, Wenxuan Zhou, Pradyot Prakash, Di Jin, Yuning Mao, Quintin Fettes, Arya Talebzadeh, Sinong Wang, Han Fang, Carolyn Rose, Daniel Fried, and Hejia Zhang
Annual Meeting of the Association for Computational Linguistics (ACL), 2025

24. **Agent Workflow Memory**
Zora Zhiruo Wang, Jiayuan Mao, Daniel Fried, and Graham Neubig
ICML, 2025

25. **Dynamic Coalition Structure Detection in Natural Language-based Interactions**
Abhishek N. Kulkarni*, Andy Liu*, Jean-Raphael Gaglione, Daniel Fried, and Ufuk Topcu
International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2025

26. **AutoPresent: Designing Structured Visuals from Scratch**
Jiaxin Ge*, Zora Zhiruo Wang*, Xuhui Zhou, Yi-Hao Peng, Sanjay Subramanian, Qinyue Tan, Maarten Sap, Alane Suhr**, Daniel Fried**, Graham Neubig**, and Trevor Darrell**
Conference on Computer Vision and Pattern Recognition (CVPR), 2025

27. **CRScore: Grounding Automated Evaluation of Code Review Comments in Code Claims and Smells**
Atharva Naik, Marcus Alenius, Daniel Fried, and Carolyn Rose
North American Chapter of the Association for Computational Linguistics (NAACL), 2025

28. **CodeRAG-Bench: Can Retrieval Augment Code Generation?**
Zora Zhiruo Wang*, Akari Asai*, Xinyan Velocity Yu, Frank F. Xu, Yiqing Xie, Graham Neubig, and Daniel Fried
Findings of NAACL, 2025

29. **BigCodeBench: Benchmarking Code Generation with Diverse Function Calls and Complex Instructions**
Terry Yue Zhuo et al. (33 authors from the BigCode project)
International Conference on Learning Representations (ICLR), 2025

30. **Human-Aligned Chess with a Bit of Search**
Yiming Zhang, Athul Paul Jacob, Vivian Lai, Daniel Fried, and Daphne Ippolito
International Conference on Learning Representations (ICLR), 2025

31. **Repetition Improves Language Model Embeddings**
Jacob Mitchell Springer, Suhas Kotha, Daniel Fried, Graham Neubig, and Aditi Raghunathan
International Conference on Learning Representations (ICLR), 2025

32. **Dissecting Adversarial Robustness of Multimodal LM Agents**
Chen Henry Wu, Rishi Shah, Jing Yu Koh, Ruslan Salakhutdinov, Daniel Fried, and Aditi Raghunathan
International Conference on Learning Representations (ICLR), 2025

33. **Comparative Knowledge Distillation**
Alex Tianyi Xu*, Alex Wilf*, Paul Pu Liang, Alexander Obolenskiy, Daniel Fried, and Louis-Philippe Morency
Winter Conference on Applications of Computer Vision (WACV), 2024

34. **ECCO: Can We Improve Model-Generated Code Efficiency Without Sacrificing Functional Correctness?**
Siddhant Waghjale*, Vishruth Veerendranath*, Zora Zhiruo Wang, and Daniel Fried
Empirical Methods in Natural Language Processing (EMNLP), 2024

35. **What Are Tools Anyway? A Survey from the Language Model Perspective**
Zora Zhiruo Wang, Zhoujun Cheng, Hao Zhu, Daniel Fried, and Graham Neubig
Conference on Language Modeling (COLM), 2024

36. **Human-Agent Cooperation in Games under Incomplete Information through Natual Language Communication**
Shenghui Chen, Daniel Fried, and Ufuk Topcu
International Joint Conference on Artificial Intelligence (IJCAI), 2024

37. **Evaluating Large Language Model Biases in Persona-Steered Generation**
Andy Liu, Mona T. Diab, and Daniel Fried
Findings of ACL, 2024

38. **Is the Pope Catholic? Yes, the Pope is Catholic. Generative Evaluation of Intent Resolution in LLMs**

Akhila Yerukola, Saujas Vaduguru, Daniel Fried, and Maarten Sap
Annual Meeting of the Association for Computational Linguistics (ACL), 2024

39. **VisualWebArena: Evaluating Multimodal Agents on Realistic Visual Web Tasks**
Jing Yu Koh, Robert Lo*, Lawrence Jang*, Vikram Duvvur*, Ming Chong Lim*, Po-Yu Huang*, Graham Neubig, Shuyan Zhou, Ruslan Salakhutdinov, and Daniel Fried
Annual Meeting of the Association for Computational Linguistics (ACL), 2024

40. **TroVE: Inducing Verifiable and Efficient Toolboxes for Solving Programmatic Tasks**
Zhiruo Wang, Graham Neubig, and Daniel Fried
International Conference on Machine Learning (ICML), 2024

41. **Amortizing Pragmatic Program Synthesis with Rankings**
Yewen Pu, Saujas Vaduguru, Priyan Vaithilingam, Elena Glassman, and Daniel Fried
International Conference on Machine Learning (ICML), 2024

42. **Asking More Informative Questions for Grounded Retrieval**
Sedrick Keh, Justin T. Chiu, and Daniel Fried
Findings of NAACL, 2024

43. **Generating Pragmatic Examples to Train Neural Program Synthesizers**
Saujas Vaduguru, Daniel Fried, and Yewen Pu
International Conference on Learning Representations (ICLR), 2024

44. **Sotopia: Interactive Evaluation for Social Intelligence in Language Agents**
Xuhui Zhou*, Hao Zhu*, Leena Mathur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, and Maarten Sap
International Conference on Learning Representations (ICLR), 2024

45. **WebArena: A Realistic Web Environment for Building Autonomous Agents**
Shuyan Zhou*, Frank Xu*, Hao Zhu**, Xuhui Zhou**, Robert Lo**, Abishek Sridhar**, Xianyi Cheng, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig
International Conference on Learning Representations (ICLR), 2024

46. **API-Assisted Code Generation for Question Answering on Varied Table Structures**
Yihan Cao*, Shuyi Chen*, Ryan Liu*, Zhiruo Wang, and Daniel Fried
Empirical Methods in Natural Language Processing (EMNLP), 2023

47. **Symbolic Planning and Code Generation for Grounded Dialogue**
Justin Chiu, Wenting Zhao, Derek Chen, Saujas Vaduguru, Alexander Rush, and Daniel Fried
Empirical Methods in Natural Language Processing (EMNLP), 2023

48. **Pragmatics in Language Grounding: Phenomena, Tasks, and Modeling Approaches**
Daniel Fried*, Nicholas Tomlin*, Jennifer Hu, Roma Patel, and Aida Nematzadeh
Findings of EMNLP, 2023

49. **Execution-Based Evaluation for Open-Domain Code Generation**
Zhiruo Wang, Shuyan Zhou, Daniel Fried, and Graham Neubig
Findings of EMNLP, 2023

50. **Data Augmentation for Code Translation with Comparable Corpora and Multiple References**
Yiqing Xie, Atharva Naik, Daniel Fried, Carolyn Rose
Findings of EMNLP, 2023

51. **AutoReply: Detecting Nonsense in Dialogue Introspectively with Discriminative Replies**
Weiyan Shi, Emily Dinan, Adi Renduchintala, Daniel Fried, Athul Paul Jacob, Zhou Yu, and Mike Lewis
Findings of EMNLP, 2023

52. **Generating Images with Multimodal Language Models**
Jing Yu Koh, Daniel Fried, and Ruslan Salakhutdinov
Neural Information Processing Systems (NeurIPS), 2023

53. **Pragmatic Inference with a CLIP Listener for Contrastive Captioning**
Jiefu Ou, Benno Krojer, and Daniel Fried
Findings of ACL, 2023

54. **Contrastive Decoding: Open-ended Text Generation as Optimization**
Xiang Lisa Li, Ari Holtzman, Daniel Fried, Percy Liang, Jason Eisner, Tatsunori Hashimoto, Luke Zettlemoyer, and Mike Lewis
Annual Meeting of the Association for Computational Linguistics (ACL), 2023

55. **Grounding Language Models to Images for Multimodal Inputs and Outputs**
Jing Yu Koh, Ruslan Salakhutdinov, and Daniel Fried
International Conference on Machine Learning (ICML), 2023

56. **Coder Reviewer Reranking for Code Generation**
Tianyi Zhang, Tao Yu, Tatsunori B. Hashimoto, Mike Lewis, Wen-tau Yih, Daniel Fried, and Sida I. Wang
International Conference on Machine Learning (ICML), 2023

57. **DS-1000: A Natural and Reliable Benchmark for Data Science Code Generation**
Yuhang Lai*, Chengxi Li*, Yiming Wang*, Tianyi Zhang*, Ruiqi Zhong*, Luke Zettlemoyer, Scott Wen-tau Yih, Daniel Fried, Sida I. Wang, and Tao Yu
International Conference on Machine Learning (ICML), 2023

58. **InCoder: A Generative Model for Code Infilling and Synthesis**
Daniel Fried*, Armen Aghajanyan*, Jessy Lin, Sida I. Wang, Eric Wallace, Freda Shi, Ruiqi Zhong, Wen-tau Yih, Luke Zettlemoyer, and Mike Lewis
International Conference on Learning Representations (ICLR), 2023

59. **Natural Language to Code Translation with Execution**
Freda Shi, Daniel Fried, Marjan Ghazvininejad, Luke Zettlemoyer, and Sida I. Wang
Empirical Methods in Natural Language Processing (EMNLP), 2022

60. **Neural Theory-of-Mind? On the Limits of Social Intelligence in Large LMs**
Maarten Sap, Ronan Le Bras, Daniel Fried, and Yejin Choi
Empirical Methods in Natural Language Processing (EMNLP), 2022

61. **G3: Geolocation via Guidebook Grounding**
Grace Luo*, Giscard Biamby*, Trevor Darrell, Daniel Fried, and Anna Rohrbach
Findings of EMNLP, 2022

62. **Inferring Rewards from Language in Context**
Jessy Lin, Daniel Fried, Dan Klein, and Anca Dragan
Annual Meeting of the Association for Computational Linguistics (ACL), 2022

63. **Reference-Centric Models for Grounded Collaborative Dialogue**
Daniel Fried, Justin Chiu, and Dan Klein
Empirical Methods in Natural Language Processing (EMNLP), 2021

64. **Modular Networks for Compositional Instruction Following**
Rodolfo Corona, Daniel Fried, Coline Devin, Dan Klein, and Trevor Darrell
North American Chapter of the Association for Computational Linguistics (NAACL), 2021

65. **Learning to Segment Actions from Observation and Narration**
Daniel Fried, Jean-Baptiste Alayrac, Phil Blunsom, Chris Dyer, Stephen Clark, Aida Nematzadeh
Annual Meeting of the Association for Computational Linguistics (ACL), 2020

66. **Cross-Domain Generalization of Neural Constituency Parsers**
Daniel Fried*, Nikita Kitaev*, and Dan Klein
Annual Meeting of the Association for Computational Linguistics (ACL), 2019

67. **Are You Looking? Grounding to Multiple Modalities in Vision-and-Language Navigation**
Ronghang Hu, Daniel Fried, Anna Rohrbach, Dan Klein, Trevor Darrell, and Kate Saenko
Annual Meeting of the Association for Computational Linguistics (ACL), 2019

68. **Pragmatically Informative Text Generation**
Sheng Shen, Daniel Fried, Jacob Andreas, and Dan Klein
North American Chapter of the Association for Computational Linguistics (NAACL), 2019

69. **Speaker-Follower Models for Vision-and-Language Navigation**
Daniel Fried*, Ronghang Hu*, Volkan Cirik*, Anna Rohrbach, Jacob Andreas, Louis-Philippe Morency,

Taylor Berg-Kirkpatrick, Kate Saenko, Dan Klein**, and Trevor Darrell**
Neural Information Processing Systems (NeurIPS), 2018

70. **Policy Gradient as a Proxy for Dynamic Oracles in Constituency Parsing**
Daniel Fried and Dan Klein
Annual Meeting of the Association for Computational Linguistics (ACL), 2018

71. **Unified Pragmatic Models for Generating and Following Instructions**
Daniel Fried, Jacob Andreas, and Dan Klein
North American Chapter of the Association for Computational Linguistics (NAACL), 2018

72. **Effective Inference for Generative Neural Parsing**
Mitchell Stern, Daniel Fried, and Dan Klein
Empirical Methods in Natural Language Processing (EMNLP), 2017

73. **Improving Neural Parsing by Disentangling Model Combination and Reranking Effects**
Daniel Fried*, Mitchell Stern*, and Dan Klein
Annual Meeting of the Association for Computational Linguistics (ACL), 2017

74. **Towards Using Social Media to Identify Individuals at Risk for Preventable Chronic Illness**
Dane Bell, Daniel Fried, Luwen Huangfu, Mihai Surdeanu, and Stephen Kobourov
Language Resources and Evaluation Conference (LREC), 2016

75. **Low-Rank Tensors for Verbs in Compositional Distributional Semantics**
Daniel Fried, Tamara Polajnar, and Stephen Clark
Annual Meeting of the Association for Computational Linguistics (ACL), 2015

76. **Analyzing the Language of Food on Social Media**
Daniel Fried, Mihai Surdeanu, Stephen Kobourov, Melanie Hingle, and Dane Bell
International Conference on Big Data, 2014

77. **Maps of Computer Science**
Daniel Fried and Stephen Kobourov
Pacific Visualization Symposium (PacificVis), 2014

78. **Predicting Parallelization of Sequential Programs Using Supervised Learning**
Daniel Fried, Zhen Li, Ali Jannesari, and Felix Wolf
International Conference on Machine Learning and Applications, 2013

79. **A Generative Probabilistic Framework for Learning Spatial Language**
Colin Dawson, Jeremy Wright, Antons Rebguns, Marco Valenzuela Escarcega, Daniel Fried, and Paul Cohen
International Conference on Development and Learning, 2013. **Best Paper Award**

80. **Bayesian Geometric Modeling of Indoor Scenes**
Luca Del Pero, Joshua Bowdish, Daniel Fried, Bonnie Kermgard, Emily Hartley, and Kobus Barnard
Conference on Computer Vision and Pattern Recognition (CVPR), 2012

Workshop Publications

81. **SantaCoder: Don't Reach for the Stars**
Loubna Ben Allal*, Raymond Li*, Denis Kocetkov*, et al. (41 authors from the BigCode Project)
Deep Learning for Code Workshop, 2023. **Best Paper Award**

82. **Modeling Perspective-Dependent Ambiguity in Grounded Collaborative Dialogue**
Justin Chiu, Wenting Zhao, Alexander M. Rush, and Daniel Fried
Wordplay: When Language Meets Games Workshop, 2022

83. **Interactive Assignments for Teaching Structured Neural NLP**
David Gaddy, Daniel Fried, Nikita Kitaev, Mitchell Stern, Rodolfo Corona, John DeNero, and Dan Klein
Teaching NLP Workshop at NAACL, 2021

84. **Challenges for Using Social Media for Early Detection of Type II Diabetes Mellitus**
Dane Bell, Daniel Fried, Luwen Huangfu, Mihai Surdeanu, and Stephen Kobourov
International Workshop on Social Media World Sensors, 2016

85. **Learning Low-Rank Tensors for Transitive Verbs**
Daniel Fried, Tamara Polajnar, and Stephen Clark
Advances in Distributional Semantics Workshop, 2015

86. **Incorporating both Distributional and Relational Semantics in Word Representations**
Daniel Fried and Kevin Duh
International Conference on Learning Representations (ICLR) Workshop, 2015

PREPRINTS

87. **Predicting Alignment Generalization with Value Representations**
Andy Liu, Mehar Bhatia, Karolina Stanczak, Mona Diab, Vered Shwartz, and Daniel Fried
arXiv, 2026

88. **CUA-Speedrun: Standardized Benchmarking of the Speed of Computer-Use Agents**
Pranjal Aggarwal*, Lawrence Keunho Jang*, Sean Welleck, Daniel Fried, Ruslan Salakhutdinov, Jing Yu Koh*
preprint, 2026

89. **Efficient Test-Time Adaptation through Human-AI Interaction**
Zora Zhiruo Wang, Apurva Gandhi, Rulin Shao, Aspen Chen, Jonas Mueller, Zhiqi Liang, Jett Chen, Michael Ryan, Qianou Ma, Luxi He, Zhoujun Cheng, Andre He, Seungone Kim, Jiayi Geng, Mingqian Zheng, Weiwei Sun, Zheyuan Zhang, Xinran Zhao, Yike Wang, Abe Hou, Liwei Jiang, Pang Wei Koh, Diyi Yang, Graham Neubig, Daniel Fried
arXiv, 2026

90. **Multi-Agent Computer Use**
Jing Yu Koh, Ruslan Salakhutdinov, Daniel Fried
arXiv, 2026

91. **How Well Does Agent Development Reflect Real-World Work?**
Zora Zhiruo Wang, Sanidhya Vijayvargiya, Aspen Chen, Hanmo Zhang, Venu Arvind Arangarajan, Jett Chen, Valerie Chen, Diyi Yang, Daniel Fried, Graham Neubig
arXiv, 2026

92. **Position: Humans are Missing from AI Coding Agent Research**
Zora Zhiruo Wang*, John Yang*, Kilian Lieret*, Alexa Tartaglini, Valerie Chen, Yuxiang Wei, Zijian Wang, Lingming Zhang, Karthik Narasimhan, Ludwig Schmidt, Graham Neubig, Daniel Fried, Diyi Yang
preprint, 2026

93. **Measuring Fine-Grained Negotiation Tactics of Humans and LLMs in Diplomacy**
Wenkai Li*, Lynnette Hui Xian Ng*, Andy Liu, Daniel Fried
arXiv, 2025

94. **How Do AI Agents Do Human Work? Comparing AI and Human Workflows Across Diverse Occupations**
Zora Zhiruo Wang, Yijia Shao, Omar Shaikh, Daniel Fried, Graham Neubig, Diyi Yang
arXiv, 2025

95. **Analyzing Information Sharing and Coordination in Multi-Agent Planning**
Tianyue Ou, Saujas Vaduguru, Daniel Fried
arXiv, 2025

96. **CodeBenchGen: Creating Scalable Execution-based Code Generation Benchmarks**
Yiqing Xie, Alex Xie, Divyanshu Sheth, Pengfei Liu, Daniel Fried, and Carolyn Rose
arXiv, 2024

*,**: equal contribution

STUDENTS

PhD thesis advisor:

- Andre He (with Sean Welleck)
- Jing Yu (JY) Koh (with Ruslan Salakhutdinov)
- Andy Liu (with Mona Diab)

- Saujas Vaduguru
- Zhiruo (Zora) Wang (with Graham Neubig)

MLT thesis advisor:
- Alex Xie (with Matt Gormley and Vincent Hellendoorn)

PhD committee member:
- Shuyan Zhou
- Aman Madaan
- Ta-Chung Chi
- Nikitha Rao
- Frank Xu
- Ritam Dutt
- Ruohong Zhang
- So Yeon (Tiffany) Min
- Anthony Sicilia (Northeastern)
- Kush Jain
- Brendon Boldt
- Nicholas Tomlin (UC Berkeley)
- Sanjay Subramanian (UC Berkeley)
- Liangke Gui
- Xing Han Liu (McGill)
- Xuhui Zhou
- Yiqing Xie
- Alex Wilf
- Yuxiang Wei (UIUC)
- Akhila Yerukola

Teaching

Instructor: **11-768, AI Agents**, CMU, 2026 – present

Instructor: **11-891, Neural Code Generation**, CMU, 2024 – present

Instructor: **11-777, Multimodal Machine Learning**, CMU, 2023 – present

Instructor: **11-877, Advanced Multimodal Machine Learning**, CMU, Spring 2024

Instructor: **11-711, Advanced NLP**, CMU, Fall 2023

Instructor: **CS188, Artificial Intelligence**, UC Berkeley, Summer 2018

Teaching Assistant: **CS188, Artificial Intelligence**, UC Berkeley, Fall 2017

Teaching Assistant: **CS245, Intro to Discrete Structures**, University of Arizona, Spring 2012

Teaching Assistant: **ISTA100, Great Ideas of the Information Age**, University of Arizona, Fall 2011

Project Design: **CS288, Natural Language Processing**, UC Berkeley, Spring 2020

Invited Talks

At What Cost? Evaluating Computer Use Agents Beyond Success. Microsoft Research and AIMSEC Workshop on Agent Evaluation. Spring 2026

Doing More With Less: Behavioral Efficiency in Language Model Agents. Stanford NLP Seminar. Spring 2026

Abstractions of Agent Actions. IVADO Workshop on the Safety and Capabilities of Agents 2025; Agentic AI Frontier Seminar 2026; CMU HCII Seminar 2026; UPenn NLP Seminar 2026; MIT NLP Seminar 2026

Improving NLP Systems with Speaker Listener Games. Pragmatic Reasoning in Language Models Workshop, COLM 2025.

Code Generation With and For Agents. IVADO Bootcamp Focusing the State of Agents 2025.

Improving NLP Systems with Speaker Listener Games. **Keynote Address**, Society for Computation in Linguistics (SCiL) 2025.

Inducing Functions to Improve LLM Agents. Deep Learning For Code Workshop, ICLR 2025; REALM Workshop, ACL 2025.

Planning and Inferring with LLMs for Grounded, Interactive Tasks. Forum for Artificial Intelligence, UT Austin. Fall 2024

Benchmarks and Tree Search for Multimodal LLM Agents. SpLU-RoboNLP Workshop, ACL 2024. Summer 2024

Interacting with LLMs for Grounded Tasks. NILLI Workshop, EMNLP 2023. Spring 2023

Interacting with (code) LLMs. MIT Neurosymbolic Reading Group. Spring 2023

InCoder, SantaCoder, and StarCoder: Findings from Training Open-Source Code LLMs. Bloomberg AI; GitHub Next. Spring 2023

Using Language Strategically in Context. CoRL Workshop on Strategic Multi-Agent Interactions, 2022; Johns Hopkins University, University of Pennsylvania, UT Austin. Spring 2023

Contextual Communication in Programming. LTI Future of Code Generation Seminar, 2022

Reasoning About Actions and Rewards in Language Interactions. MIT CPL, 2022. (with Jessy Lin)

Modularity in Grounded Interaction. ViGIL Workshop, NAACL 2021. (with Rudy Corona)

Learning Grounded Pragmatic Communication. University of Arizona, TTI-Chicago, University of Southern California, Purdue, Carnegie Mellon University, UC Irvine, Université de Montréal/Mila, Allen Institute for Artificial Intelligence, Facebook AI Research, Google Research, Stanford Cognition and Language Workshop. Spring 2021

Pragmatic Models for Generating and Following Grounded Instructions. Stanford NLP Seminar, University of Arizona, USC ISI. Fall 2018–Spring 2019

Academic Service

Workshop co-organizer: Workshop on Large Language Models for Agents, ICLR 2024

Workshop co-organizer: 3rd UnImplicit Workshop, EACL 2024

Workshop co-organizer: 2nd UnImplicit Workshop, NAACL 2022

Workshop co-organizer: 2nd Advances in Language and Vision Workshop, NAACL 2021

Workshop advisory board: Theory of Mind in Communicating Agents, ICML 2023

Outstanding reviewer awards: ACL 2018, NeurIPS 2019, ACL 2020, ACL 2021, ACL 2022

Senior area chair: ACL 2024; ACL 2026

Area chair: EMNLP 2022–2024, ACL 2023, ACL Rolling Review 2024–present

Ethics chair: NAACL 2024

Action Editor: TACL 2025–present

Reviewing: TACL 2022–2025; ACL Rolling Review 2021–2024; ACL 2018–2022; EMNLP 2016–2021; NAACL 2019, 2021; AACL-IJCNLP 2020; EACL 2017, 2021; NeurIPS 2019–2023; ICML 2019, 2023, 2026; ICLR 2021–2023; AKBC 2021; COLING 2018; *SEM 2016–2018; NAACL-SRW 2016, 2018; ACL-SRW 2020; IJCAI 2016; SpLU-RoboNLP 2019, 2021; NeuralGen 2019; ViGIL 2019, 2021; DeepLo 2019; ALVR 2020, 2021; LaReL 2020; Cooperative AI 2021; Meaning in Context 2021